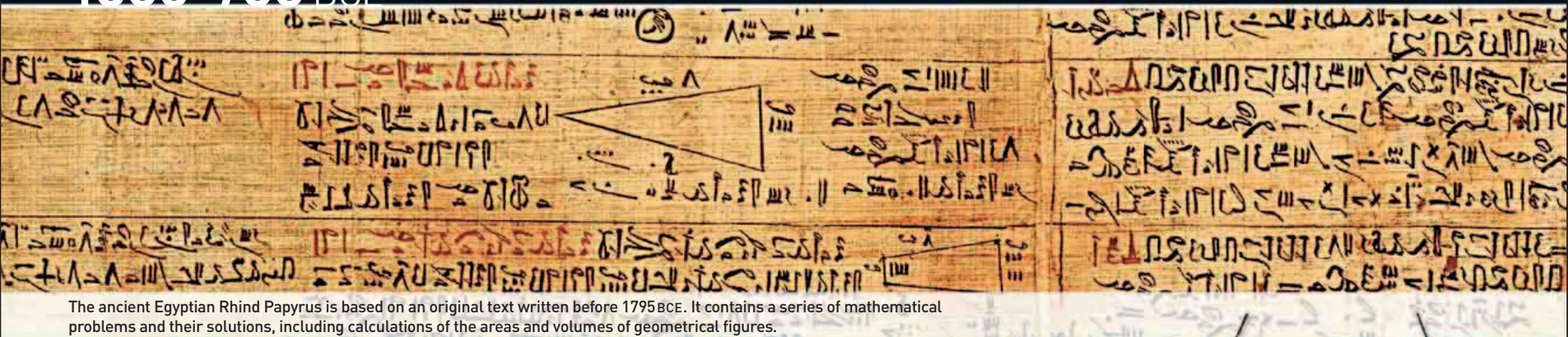




The ancient Egyptian Rhind Papyrus is based on an original text written before 1795 BCE. It contains a series of mathematical problems and their solutions, including calculations of the areas and volumes of geometrical figures.

IN THE EARLY 2ND MILLENNIUM

BCE, the composite bow was developed, probably in the steppes of Central Asia. Unlike self bows made of a single piece of wood, the **composite bow** was made of laminated strips of horn, wood, and sinew, which together provided greater range and penetration, and allowed the bow to be smaller and easier to use on horseback. The bow was **further modified to become recurved**, with the ends curving forward, which added even more strength. Composite bows spread from the

“ANOTHER REMEDY FOR SUFFERING IN HALF THE HEAD. THE SKULL OF A **CATFISH**, FRIED IN OIL. **ANOINT** THE HEAD THEREWITH.”

Ebers Papyrus 250, Egyptian medical treatise, c.1555 BCE

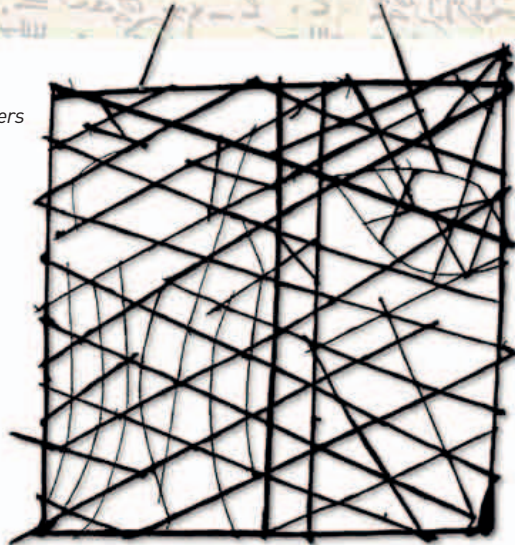
Steppes to China, where they were used during the Shang (1766–1126 BCE) and Zhou (1126–256 BCE) dynasties, and west into Egypt and Mesopotamia. There is evidence that **doctors existed in Egypt** during the Old Kingdom (c.2700–2200 BCE) and

depictions of surgery have been found on temple walls, but most knowledge of ancient Egyptian medicine comes from papyri written around 1550 BCE. These show that medicine had moved beyond a belief that disease was a divine punishment. The Edwin Smith Papyrus (c.1600 BCE) contains **details of human anatomy**, shows awareness of the link between the pulse and heartbeat, and also gives instructions for the diagnosis and **treatment of a range of ailments and injuries**. The Ebers Papyrus (c.1555 BCE), dating from about the same time, includes descriptions of diseases, tumors, and even of mental disorders such as depression.

The earliest intentional **production of iron** was in Anatolia in Turkey, which was exporting small quantities of iron by the 19th century BCE. At first, iron was smelted only on a small scale, but by 700 BCE production was widespread in Europe. Smelting also developed independently in a number of places, including Africa and

Stick chart

Made by Marshall Islanders in Micronesia, this chart uses sticks to represent currents and waves, a technique that may have been passed down from ancient Polynesians.



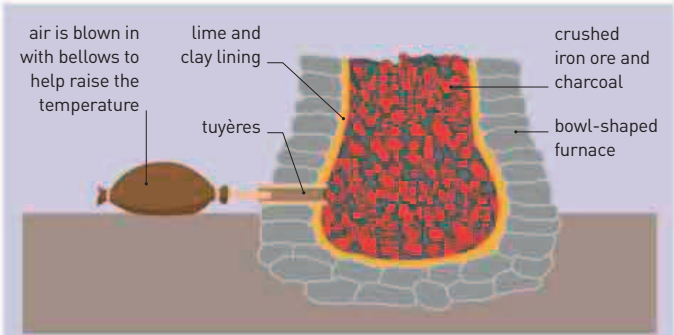
India, where the earliest evidence of ironworking is thought to date from around 1300 BCE.

Until medieval times, smelting in the West produced only bloom that needed to be hammered to remove impurities. It was only in China that furnaces capable of melting iron were developed and iron could be cast. **Evidence of cast iron** production in China dates from the 9th century BCE.

In mathematics, the Babylonians had made major advances by 1800 BCE, producing **tables of reciprocals, squares, and cubes** and using them to solve algebraic problems, such as quadratic equations. Several tablets are thought to show an awareness of Pythagoras's theorem (see 700–400 BCE). The Babylonians also estimated pi to be about 3.125, close to the actual value of about 3.142. Most of what is known of ancient Egyptian mathematics comes from **mathematical texts such**

as the Rhind Papyrus. It is based on a text written before 1795 BCE and consists of a series of problems and solutions. It shows the use of unit fractions (1/n), **solutions for linear equations**, and methods for calculating the areas of triangles, rectangles, and circles. It also shows the volumes of cylinders and pyramids.

The earliest boats recovered date from before 6000 BCE, but **early navigation was not sophisticated**. The most effective navigators of this period were the Lapita people of the Pacific (ancestors of the Polynesians), who from 1200 BCE expanded eastward to Vanuatu, New Caledonia, Samoa, and Fiji. Their voyage to Fiji involved a 530 miles (850 km) journey across open sea. To accomplish



SMELTING

Pure iron melts at 2800°F (1540°C), higher than early technology could achieve, so instead it was smelted by reducing iron ore with charcoal at around 2200°F (1200°C). The ore was packed with charcoal in bowl furnaces, and tuyères (clay nozzles) were used to blow air in to raise the temperature. The resulting molten metal was cooled to form a “bloom,” a solid lump containing iron and various impurities, which was then hammered repeatedly to remove the impurities and extract the iron.

c.1800 BCE Babylonians produce mathematical tables that include solutions for quadratic equations

c.1795 BCE Egyptians produce Rhind mathematical papyrus

c.1600 BCE Large-scale glass production begins in Egypt and the Near East

1700–1600 BCE Development of proto-Canaanite alphabetic script

c.1600 BCE Egyptians produce Edwin Smith Papyrus, containing details of human anatomy

c.1555 BCE Egyptians produce Ebers Papyrus, a description of medical conditions

c.1500 BCE Significant advances start to be made in the development of the harness

1300–1200 BCE Development of Ugaritic alphabetic script

c.1300 BCE Iron smelting develops independently in India

c.1200 BCE Mycenaeans develop enamelling and glass cloisonné inlays

c.1200 BCE Lapita people begin to make long sea voyages across the Pacific Ocean